

SQL queries

Revise 360 lesson plan — 60 minutes

Unit	Beyond OCR J277: a Python pathway supporting 2.1, 2.2 and 2.3
Lesson	SQL queries (lesson 13 of the unit)
Lesson ID	pr-l13
Experience ID	pr-l13
Plan ID	PLAN-PR-L13
Content version	6068f8dcbe81
Website	https://revise360.co.uk/experience.html?id=pr-l13

Big question, outcomes and success criteria

Big question — Which clause of a query decides which fields come back, and which clause decides which records?

Prerequisites

The lesson before this one is pr-l12. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pr-l01, pr-l02, pr-l03, pr-l04, pr-l05, pr-l06, pr-l07, pr-l08, pr-l09, pr-l10, pr-l11.

Assessable outcomes

1. **pr-l13-o1** — know how SELECT and FROM choose the fields and the table a query reads
2. **pr-l13-o2** — understand how WHERE decides which records a query hands back
3. **pr-l13-o3** — know how to join conditions with AND and OR, and how to put results in order with ORDER BY

Success criteria the pupil can check against

Taken from the briefs the independent activities themselves show, so these are the words the pupil sees. Their work has met the outcome when it does these things.

- Each line you type becomes one row of the table. **pr-l13-s1-a3**
- The word "END" tells the program the table is full. **pr-l13-s1-a3**
- Display each row the query finds on its own line. **pr-l13-s1-a5**
- Replace ____ with the name of one column. **pr-l13-s1-a6**
- Display each row on its own line, the two values separated by one space. **pr-l13-s1-a6**
- Count the rows as the query hands them back, rather than typing a number in. **pr-l13-s1-a8**
- Do not change anything above the last comment. **pr-l13-s2-a5**
- Keep only the games whose kind is card. **pr-l13-s2-a7**

Vocabulary, resources and preparation

Key vocabulary — SQL, Record and field, WHERE, ORDER BY

Resources

SLIDES	presentations/PR_L13_SQLQueries_Lesson.pptx
WORKSHEET	worksheets/PR_L13_SQLQueries_Worksheet.docx

ONLINE EXPERIENCE	experience.html?id=pr-113
CODE BANK	tools/codebank/pr-113.json

Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the Python runtime and the 360 image are already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>PR_L13_SQLQueries_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong before moving on.	Correct their own starter.	A correction written, not just a tick.
8-16	Explain and model	The new idea, then the worked example, typed live rather than shown finished.	Watch the example being built, and say what each line does.	They can name the line that does the job, not just copy it.
16-19	Set up and settle	Sign-in, launch, and what to record on paper before anyone types.	Sign in, open the lesson, read the task before typing.	Everyone is at station 1 activity 1, not typing yet.
19-48	Online: work the stations in order	Pupils work; you circulate and read the editor over shoulders.	Work the stations in order, running and checking as they go.	The station badges fill in; the editor shows their own program, not the starter.
48-53	Check understanding	The hinge question, hands down, then the reteach if the class splits.	Answer the hinge question, hands down.	The split across the options, before any reteach.
53-58	Apply or extend	The challenge activity, or the exam practice on the worksheet.	The Challenge activity, or the exam practice.	worksheet, Exam practice a) to c)

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
58-60	Exit and set up the next lesson	The exit question and what to finish before next time.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. SELECT and FROM

- SELECT names the fields you want, and FROM names the table they live in.
- SELECT * means every field, and with no WHERE every record comes back.
- Common mistake: listing the fields in the wrong order, since results come out in the order you asked for.

Challenge on this wall: Which part of a query decides how many records come back?

2. WHERE

- WHERE filters the records, keeping only the ones whose condition is true.
- Text to match goes inside quotes, and numbers do not.
- Common mistake: leaving the quotes off text, so SQL looks for a field with that name instead.

Challenge on this wall: Which records come back from a query that has no WHERE at all?

3. Comparing values

- The comparisons are =, <>, >, <, >= and <=, and they work on numbers and on text.
- 'Longer than 45' is > 45, but '45 or longer' is >= 45.
- Common mistake: using > where the question says 'or more', which loses the boundary record.

Challenge on this wall: Write the WHERE clause for games lasting 45 minutes or longer.

4. AND and OR

- AND keeps a record only when both conditions hold; OR keeps it when either one does.
- Brackets make the meaning clear when AND and OR appear in the same WHERE.
- Common mistake: using AND where the question means OR, which often gives no records at all.

Challenge on this wall: Why would AND give nothing for 'under 20 minutes or over 60'?

5. Sorting results

- ORDER BY sorts the results by a field, smallest or earliest first.
- DESC reverses that, so the largest or latest comes first instead.
- Common mistake: putting ORDER BY before WHERE, when it always comes last.

Challenge on this wall: Which clause lists the longest games first?

6. Putting it together

- The clauses always come in the order SELECT, FROM, WHERE, ORDER BY.
- Build a query one clause at a time and check what comes back at every step.
- Common mistake: mixing up the quotes, so a word is read as a field name and the query finds nothing.

Challenge on this wall: Write a query for games that are not card games and last an hour or less, longest first.

The example to model

A database keeps its data in a table: one row for each thing and one column for each fact about it. SQL is a second language for asking a table a question. SELECT names the columns you want and FROM names the table they come from.

```
import sqlite3
db = sqlite3.connect(":memory:")
db.execute("CREATE TABLE games (title TEXT, kind TEXT, players INTEGER, minutes INTEGER)")
db.execute("INSERT INTO games VALUES ('Snap Dragon', 'card', 4, 15)")
```

```
db.execute("INSERT INTO games VALUES ('Clay Pigeon', 'dice', 2, 30)")
db.execute("INSERT INTO games VALUES ('Willow Gate', 'card', 6, 50)")

for row in db.execute("SELECT title FROM games"):
    print(row[0])
```

It displays **Snap Dragon / Clay Pigeon / Willow Gate**

Decisions to model out loud, not just the finished answer: say why you chose that name, why that line comes before the next one, and what you would try first if it did not work. The example deliberately uses different data from the task, so a pupil cannot copy it and be finished.

Retrieval and the check for understanding

Starter — A table called Tree holds the fields Name, Height and Native, and 30 records. Write down how many fields and how many records come back from SELECT Name, Height FROM Tree, and which clause you would add to cut the records down.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pr-113-s3-c2), so the wording a pupil sees matches the wording you say.

What will this program display?

```
import sqlite3
db = sqlite3.connect(":memory:")
db.execute("CREATE TABLE games (title TEXT, kind TEXT, players INTEGER, minutes INTEGER)")
db.execute("INSERT INTO games VALUES ('Snap Dragon', 'card', 4, 15)")
db.execute("INSERT INTO games VALUES ('Clay Pigeon', 'dice', 2, 30)")
db.execute("INSERT INTO games VALUES ('Willow Gate', 'card', 6, 50)")

for row in db.execute("SELECT title FROM games WHERE minutes > 30"):
    print(row[0])
print("---")
for row in db.execute("SELECT title FROM games WHERE minutes >= 30"):
    print(row[0])
```

Correct: Willow Gate --- Clay Pigeon Willow Gate

What a pupil might choose instead

- Willow Gate --- Willow Gate
- Clay Pigeon Willow Gate --- Clay Pigeon Willow Gate

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: > leaves the boundary value out and >= lets it in. The row sitting exactly on the number is the only one that can tell the two apart.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Misconceptions this lesson addresses

- listing the fields in the wrong order, since results come out in the order you asked for. wall panel
- leaving the quotes off text, so SQL looks for a field with that name instead. wall panel
- SQL is hunting for a column called word, because nothing tells it that word is text. pr-113-s2-a5, a Fix it activity
- using > where the question says 'or more', which loses the boundary record. wall panel

- What two kinds of value is + being asked to join? pr-113-s3-a5, a Fix it activity
- using AND where the question means OR, which often gives no records at all. wall panel
- Read the right-hand side of the OR on its own. What is 'party' being compared with? pr-113-s4-a5, a Fix it activity
- putting ORDER BY before WHERE, when it always comes last. wall panel

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	15	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	11	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.
Produce	12	The pupil writes it from the brief and nothing else.
Challenge	6	The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	SELECT and FROM	8	SQL stands for Structured Query Language, and what it is for is asking rather than...

#	STATION	ACTIVITIES	RECORD ON PAPER
2	WHERE	7	WHERE looks at each record on its own and decides, for that one record, whether to...
3	Comparing values	7	SQL writes 'not equal to' as <>, where the same test in Python is written...
4	AND and OR	7	AND narrows the records down while OR widens them out, so the one that can only ever give you fewer records is...
5	Sorting results	7	Ascending is the order ORDER BY uses unless you say otherwise, and the word that asks for the other direction is...
6	Putting it together	7	One query can filter and sort at once, which is why that work belongs in the query and not in...
final	Final challenge: build the query	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Python: the order is not negotiable. Each activity teaches what the next one needs, so a pupil works through them in order and completes every one. A shorter lesson, homework or a slow worker does not authorise skipping: the pupil stops where they are and resumes at their earliest unfinished question next time, because progress saves after every answer.

A paper activity, a help request or a tick on this plan does not mark an online question complete, and there is no teacher bypass. If a pupil is stuck after the hint, that is the moment to teach them, not to move them on.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pr-113-x1	A query is written <code>SELECT Title FROM Game ORDER BY Length WHERE Players = 2</code> . State the error in it, and write the query out corrected.	2	worksheet, Exam practice a); deck, Exam practice slide
pr-113-x2	A table called Track holds the fields Title, Artist, Seconds and Genre. Write a query that lists the title and artist of every track longer than 240 seconds, longest first.	4	worksheet, Exam practice b); deck, Exam practice slide
pr-113-x3	Using the same Track table, write a query that lists the title of every track that is Jazz or Blues and lasts under 180 seconds.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pr-113.json`, which is not published with the website.

Every programming activity in this lesson is marked by running the pupil's program against its test cases, so the mark is the program's behaviour and not its appearance. The model solutions are in `answers/codebank/pr-113.json`.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What will this program display?** — Snap Dragon Clay Pigeon Willow Gate; minutes minutes minutes
- **What will this program display?** — Clay Pigeon Snap Dragon Willow Gate; Snap Dragon Clay Pigeon Willow Gate

- **What will this program display?** — Clay Pigeon Willow Gate; card dice card
- **What will this program display?** — Clay Pigeon; Willow Gate
- **What will this program display?** — Willow Gate --- Willow Gate; Clay Pigeon Willow Gate --- Clay Pigeon Willow Gate

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- Writing a SELECT ... FROM query
- Filtering records with WHERE
- Using AND, OR and ORDER BY

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

This is the last lesson of the unit. The next retrieval of it is the unit test.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

If this lesson takes more than an hour

It will. There are 44 activities across 6 stations, and a beginner does not finish 44 of them in twenty-nine minutes. Plan it as two or three sessions of an hour: stations 1 and 2 in the first, 3 and 4 in the second, 5, 6 and the final challenge in the third. Each session keeps the same starter-explain-model opening; the online block resumes where the class got to. Honest total: two to three hours.

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

A shorter lesson still does not authorise skipping a question. The save-and-resume point is wherever the pupil got to; there is no way to mark a question complete without completing it, and no teacher bypass.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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